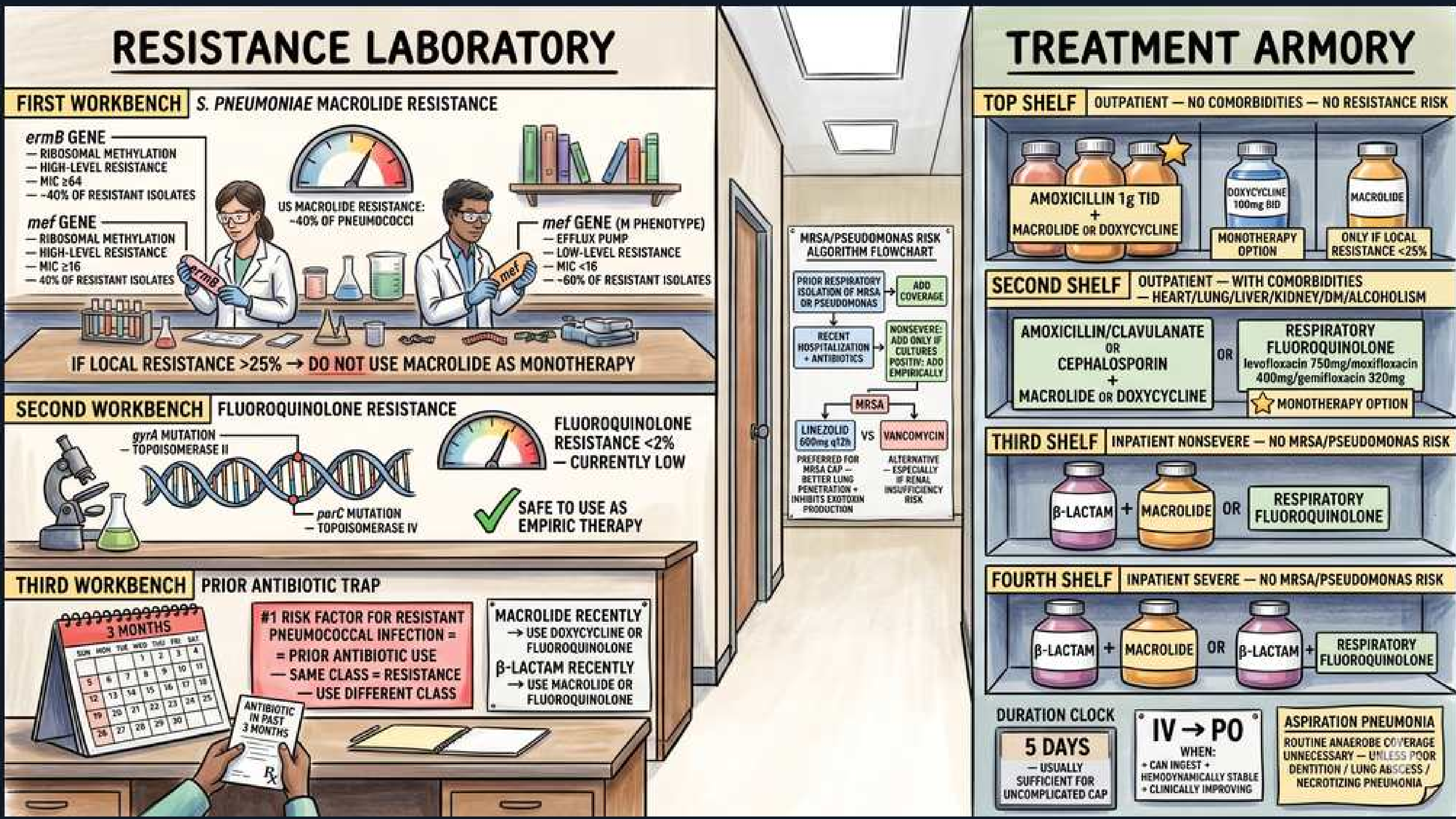
Pneumonia: Resistance Lab & CAP Treatment Armory



1. ermB macrolide resistance

WHAT YOU SEE

First workbench, ribosomal methylation

WHAT IT ENCODES

ermB causes ribosomal methylation giving high-level macrolide resistance (MIC >=16) in ~40% of pneumococci. Confers cross-resistance to macrolides, lincosamides, and streptogramin B (MLSb phenotype).

BOARD TRAP

High-level (ermB) macrolide resistance means azithromycin monotherapy fails — when local pneumococcal macrolide resistance exceeds 25%, do not use a macrolide alone.

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2. mefE efflux resistance

WHAT YOU SEE

M-phenotype, efflux pump

WHAT IT ENCODES

mefE encodes an efflux pump producing low-level macrolide resistance (MIC 1-32) in ~40% of resistant isolates. Lower-level than ermB but still clinically relevant for monotherapy failure.

BOARD TRAP

Even low-level mef efflux resistance can cause clinical failure — don't assume macrolide monotherapy is safe just because resistance is the efflux (not methylation) type.

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3. >25% rule

WHAT YOU SEE

Red warning, do-not-use-macrolide-monotherapy

WHAT IT ENCODES

If community S. pneumoniae macrolide resistance exceeds 25%, macrolides cannot be used as empiric monotherapy for CAP. This threshold drives the combination beta-lactam + macrolide regimens.

BOARD TRAP

This is the board number: 25% local resistance is the cutoff to abandon macrolide monotherapy — not 10% or 50%.

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4. Fluoroquinolone resistance

WHAT YOU SEE

Second workbench, gyrA/parC mutations

WHAT IT ENCODES

Resistance arises from gyrA (topoisomerase II/DNA gyrase) and parC (topoisomerase IV) mutations. Currently <2% in pneumococci, so respiratory fluoroquinolones remain reliable empiric therapy.

BOARD TRAP

Fluoroquinolone resistance is still rare (<2%), so a respiratory FQ is a valid single agent for CAP — but recent FQ use selects resistance, so rotate drug classes.

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5. Prior antibiotic trap

WHAT YOU SEE

Third workbench, 3-month calendar

WHAT IT ENCODES

Antibiotic use within the prior 3 months is the #1 risk factor for resistant pneumococcal infection. Always switch to a different antibiotic class than the one recently used.

BOARD TRAP

If a macrolide was used recently, switch to a beta-lactam or FQ — re-using the same class within 3 months invites resistance and treatment failure.

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6. Top shelf: healthy outpatient

WHAT YOU SEE

Treatment armory top shelf, amoxicillin 1g TID

WHAT IT ENCODES

Outpatient CAP with no comorbidities and no resistance risk: amoxicillin 1 g TID, OR doxycycline, OR a macrolide (only where resistance <25%). Monotherapy is acceptable.

BOARD TRAP

Macrolide monotherapy is only first-line for healthy outpatients in LOW-resistance areas — it is wrong for comorbid or recently-treated patients.

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7. 2nd shelf: comorbid outpatient

WHAT YOU SEE

Amox-clav/cephalosporin + macrolide/doxy or FQ

WHAT IT ENCODES

Outpatient with comorbidities (heart/lung/liver/kidney disease, diabetes, alcoholism): beta-lactam (amox-clav or cephalosporin) PLUS a macrolide or doxycycline, OR respiratory fluoroquinolone monotherapy.

BOARD TRAP

Comorbid outpatients need combination beta-lactam + macrolide OR an FQ — plain amoxicillin monotherapy is inadequate here.

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8. 3rd shelf: inpatient non-severe

WHAT YOU SEE

Beta-lactam + macrolide OR respiratory FQ

WHAT IT ENCODES

Hospitalized non-severe CAP without MRSA/Pseudomonas risk: beta-lactam (ceftriaxone/ampicillin-sulbactam) plus a macrolide, OR a respiratory fluoroquinolone alone.

BOARD TRAP

Don't add vancomycin/antipseudomonal coverage to routine inpatient CAP unless specific MRSA or Pseudomonas risk factors are present.

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9. 4th shelf: inpatient severe

WHAT YOU SEE

Beta-lactam + (macrolide OR FQ)

WHAT IT ENCODES

Severe CAP (ICU) without MRSA/Pseudomonas risk: beta-lactam PLUS either a macrolide or a respiratory fluoroquinolone. Macrolide monotherapy is never adequate for severe disease.

BOARD TRAP

In severe/ICU CAP, a beta-lactam must be combined with a macrolide or FQ — a single agent (even an FQ alone) is not recommended for severe disease.

10. MRSA/Pseudomonas algorithm

WHAT YOU SEE

Center flowchart, vancomycin/linezolid vs antipseudomonal

WHAT IT ENCODES

Add anti-MRSA (vancomycin or linezolid) for prior MRSA isolation or recent hospitalization with risk factors; add antipseudomonal beta-lactam for prior Pseudomonas, structural lung disease, or recent IV antibiotics. De-escalate by cultures.

BOARD TRAP

Empiric MRSA/Pseudomonas coverage is driven by documented prior isolation or validated risk factors — not by the old HCAP label.

11. Duration 5 days

WHAT YOU SEE

Duration clock, 5 DAYS

WHAT IT ENCODES

Uncomplicated CAP needs only 5 days of therapy provided the patient is afebrile 48-72h and clinically stable. Longer courses do not improve outcomes and increase resistance.

BOARD TRAP

Five days suffices for uncomplicated CAP — extending to 7-10 days by default is outdated; only complicated/bacteremic or specific organisms need longer.

12. IV-to-PO switch

WHAT YOU SEE

IV->PO arrow, clinically-improving

WHAT IT ENCODES

Switch from IV to oral antibiotics once the patient is hemodynamically stable, improving clinically, able to ingest, and has a functioning GI tract. Does not require full defervescence.

BOARD TRAP

Don't keep a stable, improving patient on IV antibiotics until afebrile — early IV-to-PO switch is standard and shortens length of stay.

13. Aspiration pneumonia

WHAT YOU SEE

Bottom center, aspiration box

WHAT IT ENCODES

Aspiration pneumonia is usually anaerobic/oral-flora driven; ampicillin-sulbactam or amoxicillin-clavulanate is preferred. Routine anaerobic coverage is reserved for lung abscess or empyema, not simple aspiration.

BOARD TRAP

Modern guidance: don't add clindamycin/metronidazole for every aspiration event — reserve dedicated anaerobic coverage for abscess/empyema.